

Neural Turing Machine

뉴럴 + 튜링 머신

수리과학부25 윤승현 + Chatgpt

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Cognitive Science and Linguistics

Fodor and Pylyshyn's limitation of neural networks for cognitive modeling(1988)

Variable-binding and Variable-length

“Mary spoke to John” $\leftrightarrow$ “John spoke to Mary”

(associative memory의 특성상 다음과 같은 variable-binding의 특성을 가지기 힘들다.)

Cognitive Science and Linguistics

Fodor and Pylyshyn's limitation of neural networks for cognitive modeling(1988)

Variable-binding and Variable-length

fixed length input domain으로 정적인 length

(recursive processing of variable-length structure이 인류 인지의 상징으로 여겨짐)

Recursion on Neuroscience

The Unbearable Slowness of Being: Why do we live at 10 bits/s?

Jieyu Zheng and Markus Meister
Division of Biology and Biological Engineering
California Institute of Technology
{jzzheng,meister}@caltech.edu

<https://arxiv.org/pdf/2408.10234>

정보 raw input : 1Gbit/s

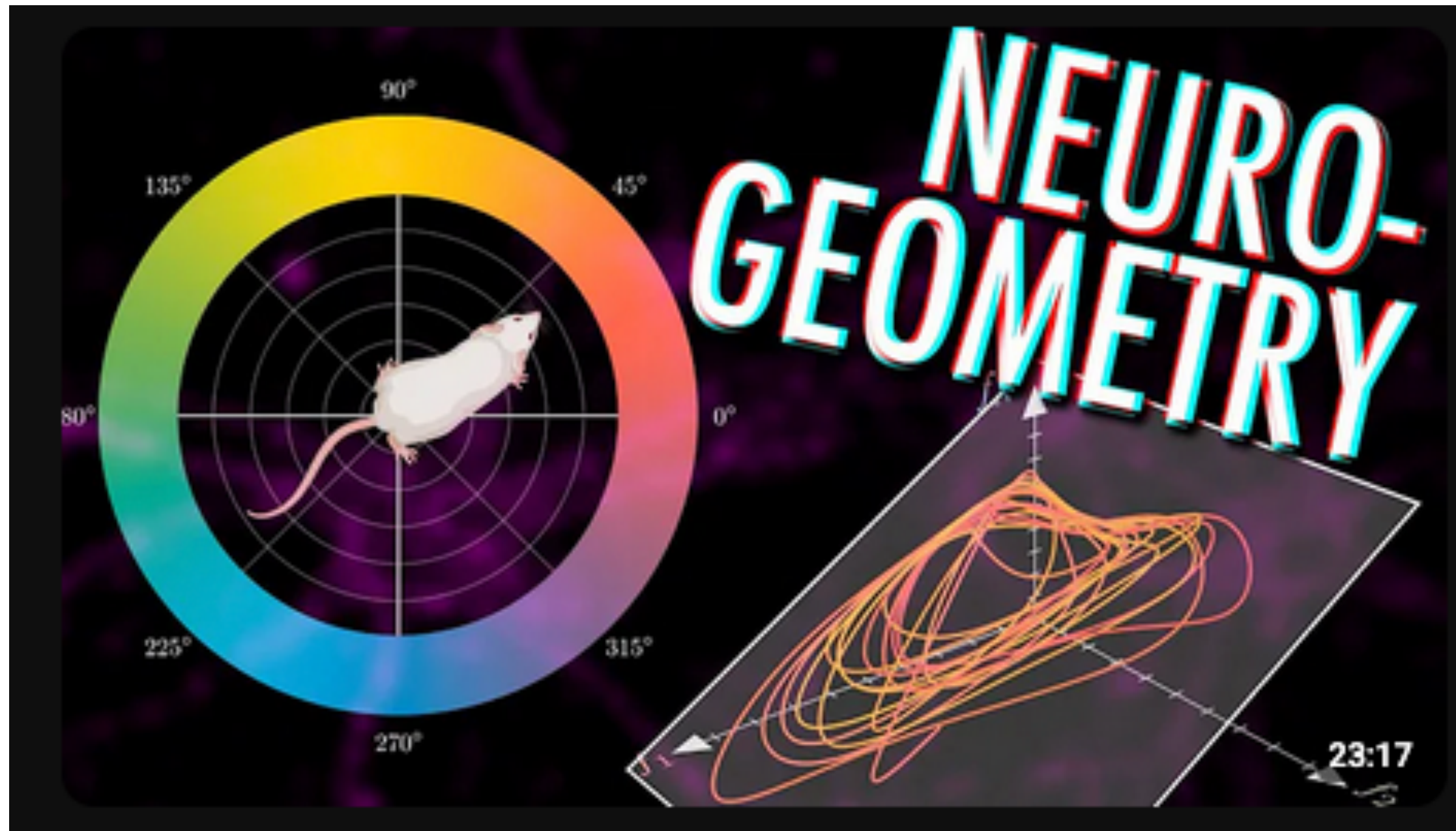
행동 output : 10bit/s (스무고개 등 사례에서)

정보의 양을 줄여나가는 연결!

(무의식의 힘)

= Convolution!!

Geometry on Neuroscience



Hebb's rule

-> Neurons that fire together, wire together.

== neurons are intertwined!

== manifold hypothesis

https://www.youtube.com/watch?v=QHj9uVmwA_0

Theory of computation

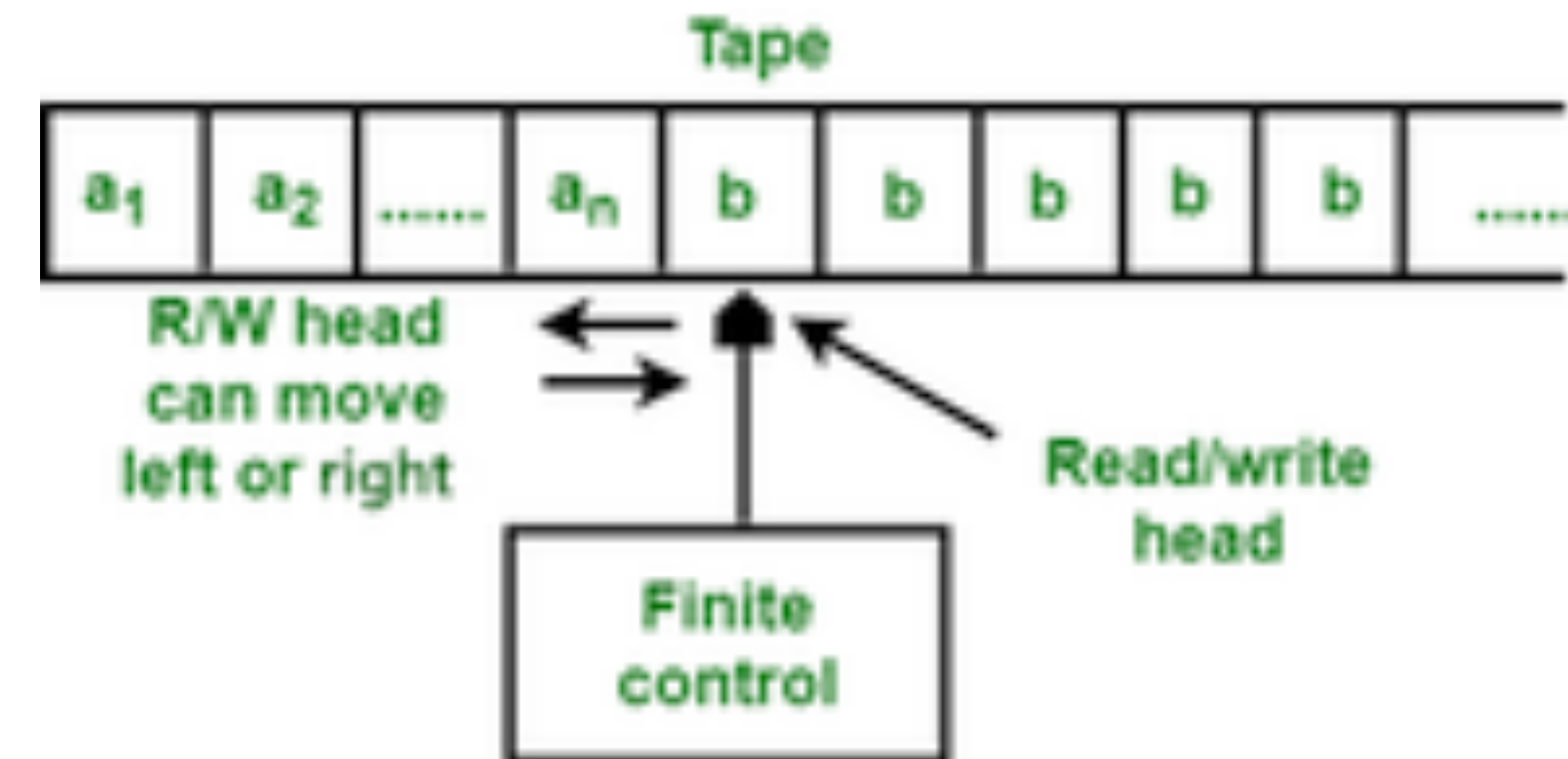
- Finite Automata (FA) : no memory
 - Pushdown Automata (PDA) : stack for limited memory
 - Linear-bounded Automata : tape is bounded compared to turing machines
 - Turing machines: unrestricted tape and head movement
- > 과연 컴퓨터는 어떤 일까지 할 수 있을까?를 알아보기 위한 컴퓨터의 수학적 모델링

Theory of computation

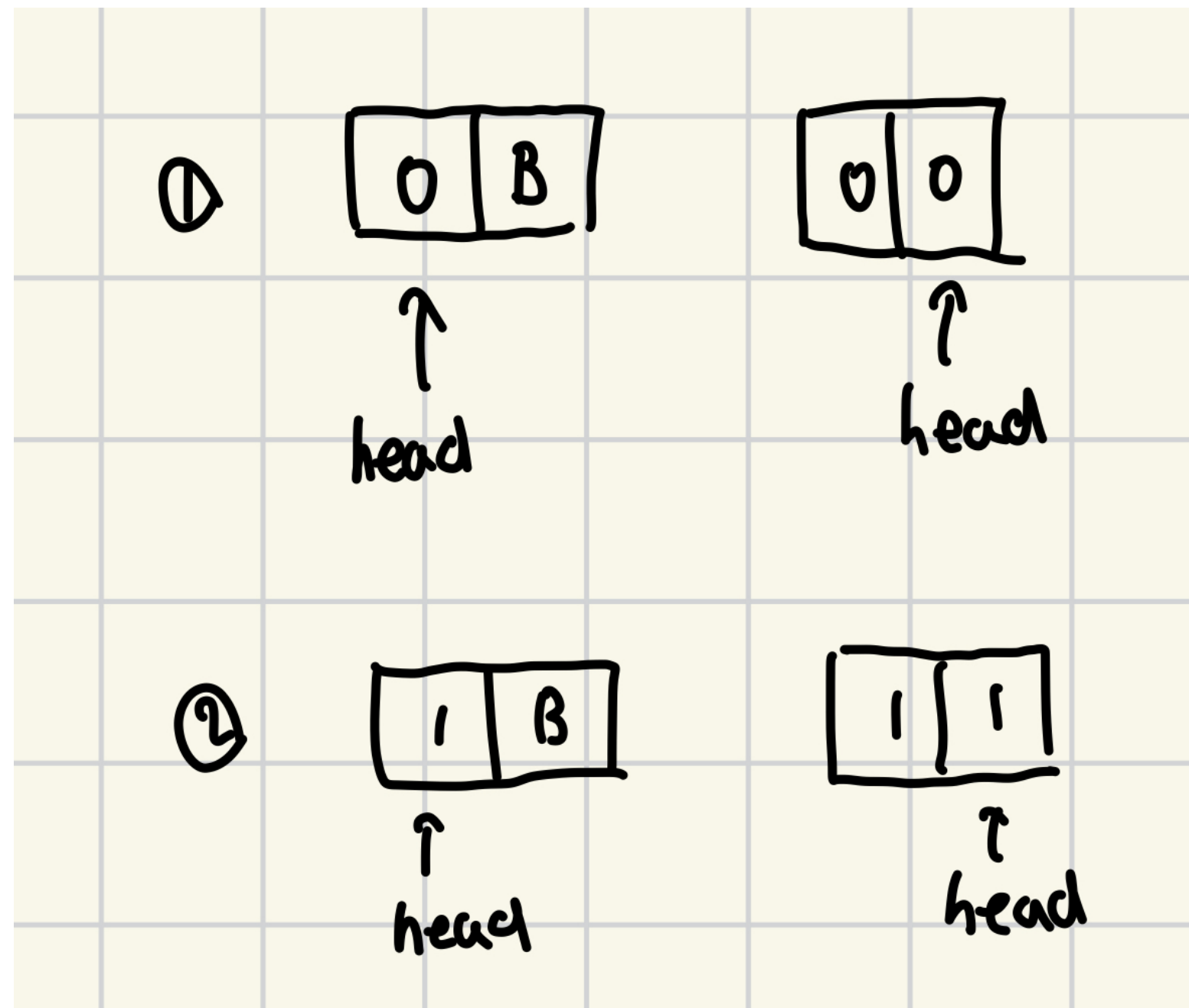
- Finite Automata (FA) : no memory -> recognizes “regular language”(regex)
 - Pushdown Automata (PDA) : stack for limited memory -> restricted programming lang
(no stackoverflow... hahaha)
 - Linear-bounded Automata : tape is bounded compared to turing machines
 - Turing machines: unrestricted tape and head movement
-> recursion(ok) variable variable-length(ok)
- > 과연 컴퓨터는 어떤 일까지 할 수 있을까?를 알아보기 위한 컴퓨터의 수학적 모델링

Turing machine

- state set
- init state
- final state set
- transition function
- (alphabets, blank symbol, tape)

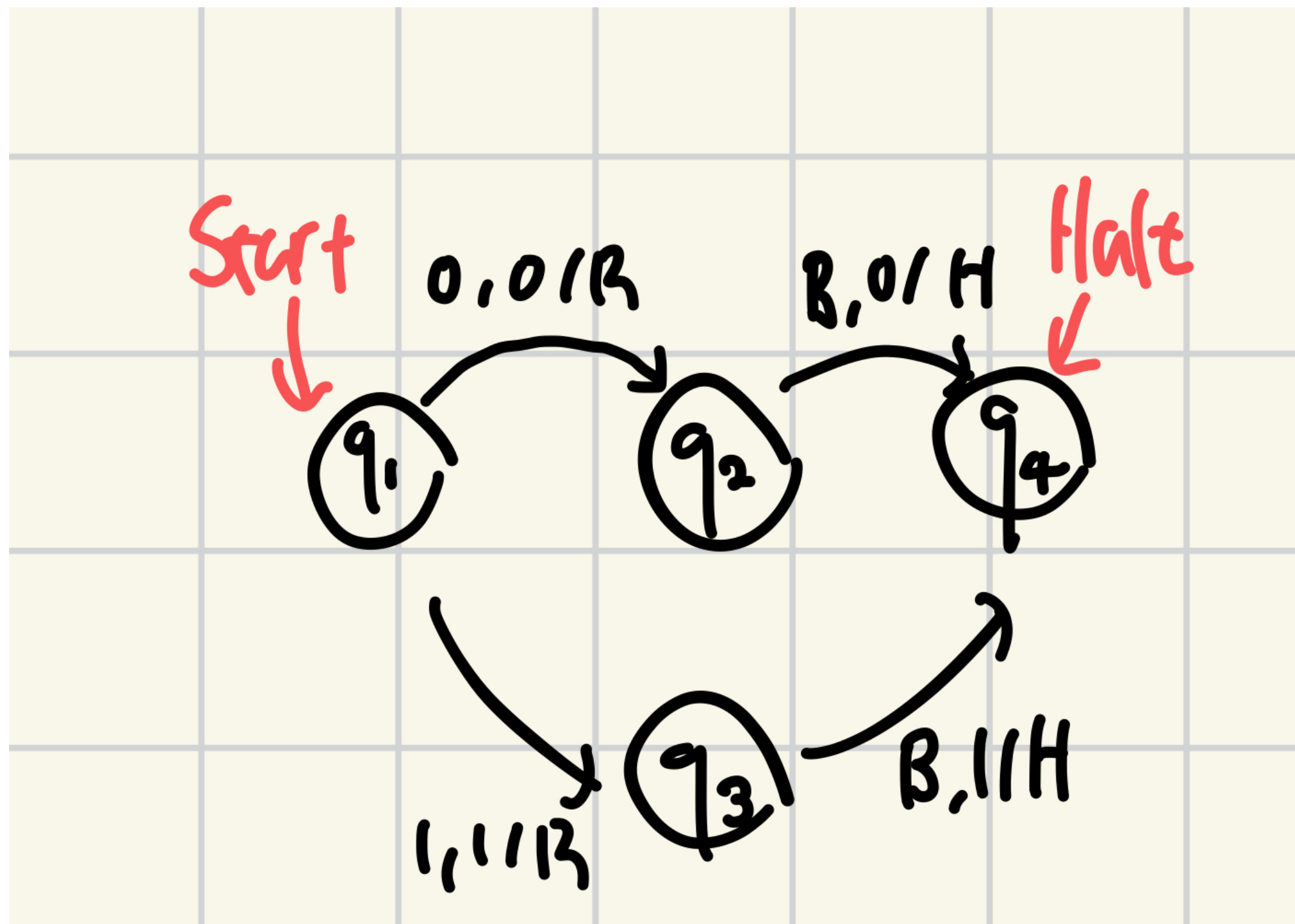


Turing machine Example - copy



오른쪽에 현재 head의 값을
복사/붙여넣기 하는 알고리즘

Turing machine Example



Deterministic Markov chain

=

$\delta(q_1, 0) = (0, 0, R)$
 $\delta(q_1, 1) = (1, 1, R)$
 $\delta(q_2, B) = (B, 0, H)$
 $\delta(q_3, B) = (B, 1, H)$

Written in function

Turing machine Example

Number of Turing machines

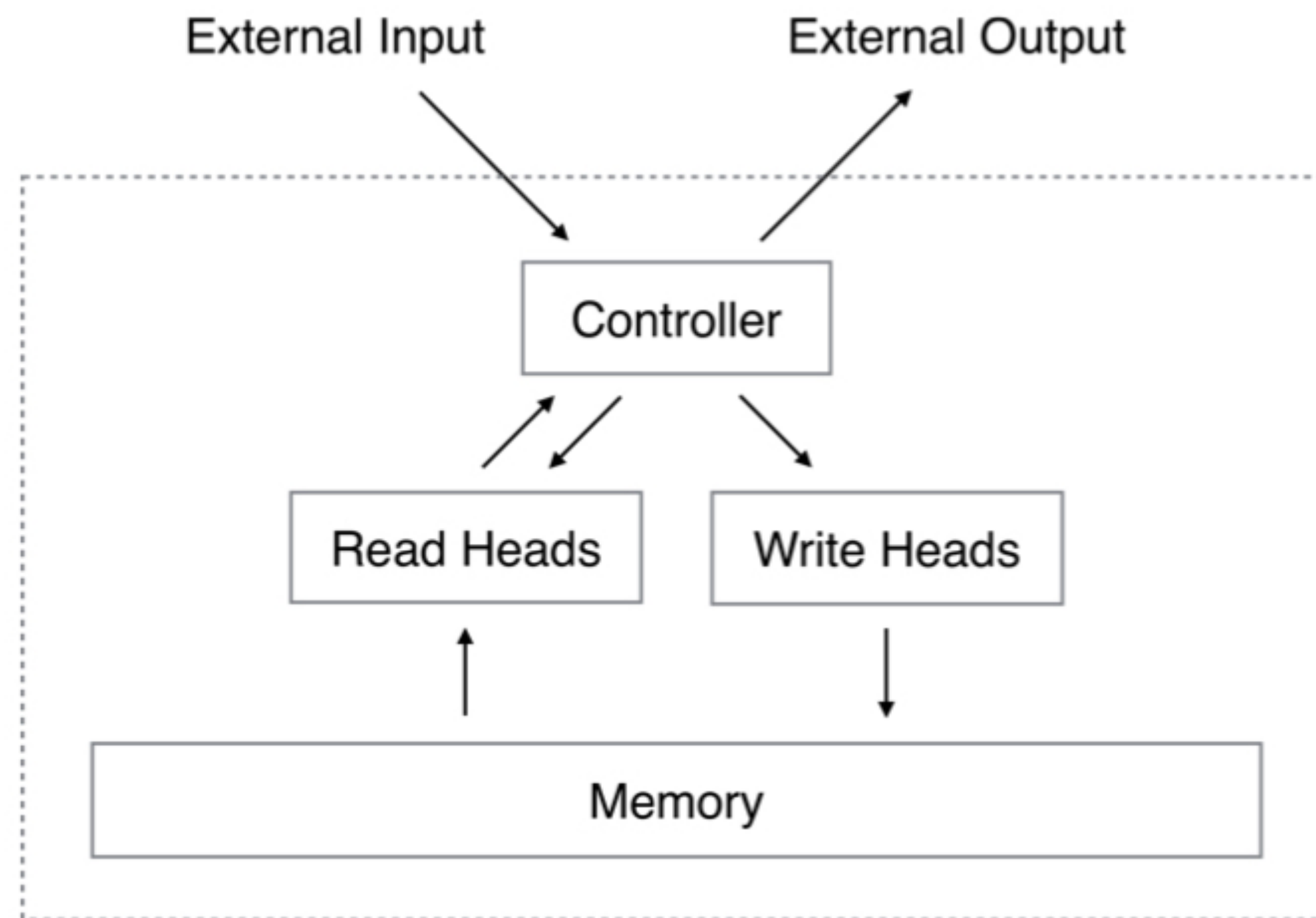
Asked 7 years, 10 months ago Modified 7 years, 9 months ago Viewed 2k times

알고리즘을 숫자로 표현할 수 있다! = 튜링 수

튜링 머신을 만드는 튜링 수도 있다!(이만큼 강력크)

So: for each of the $6 \cdot 4 = 24$ entries in the transition-table, you have $4 \cdot 2 \cdot 8 = 64$ possibilities, giving you 64^{24} possible Turing-machines using the quintuple formalization.

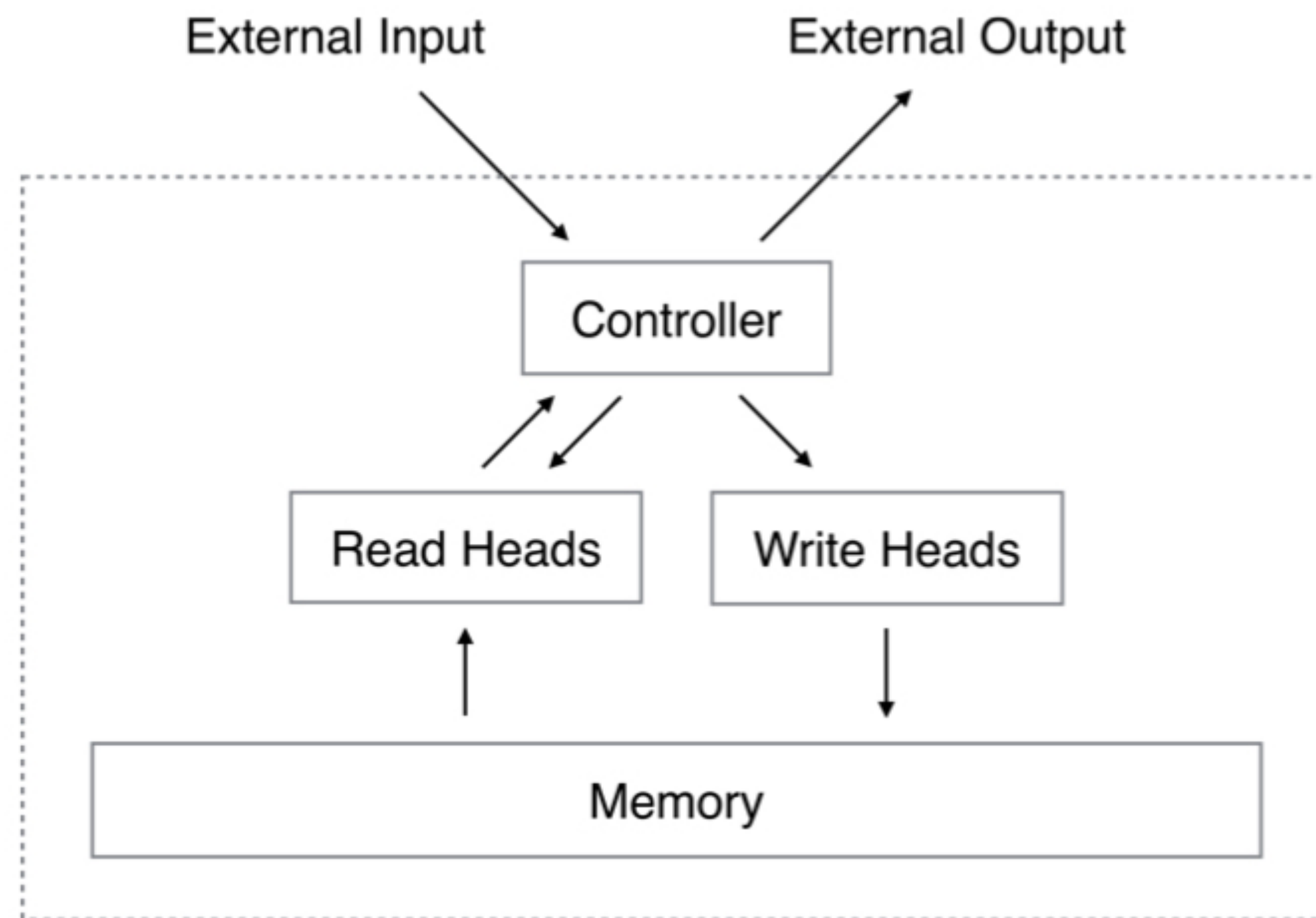
Neural Turing Machine



Controller와 Memory bank
= head와 tape

차이점은 바로!! soft-attention!

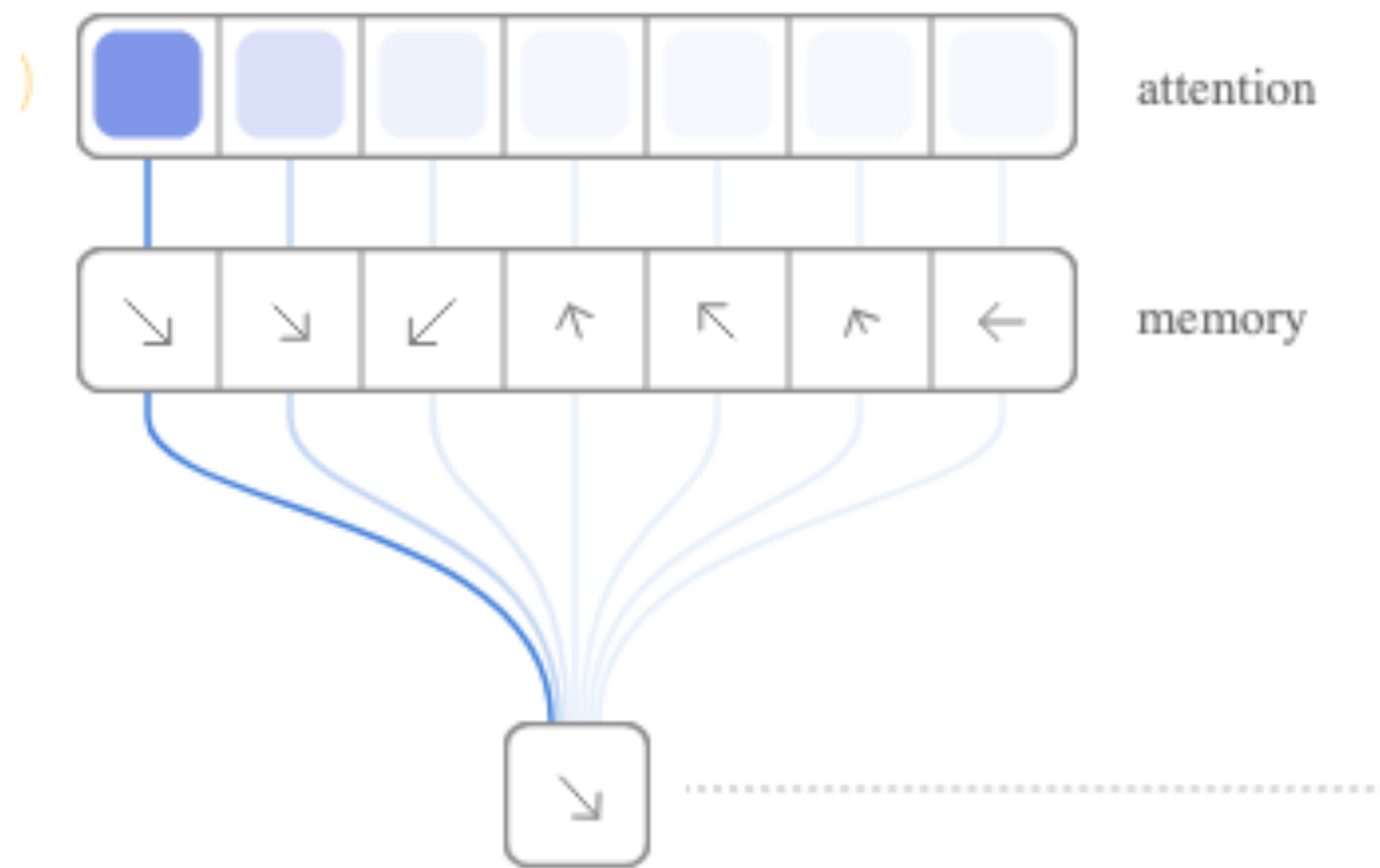
Neural Turing Machine



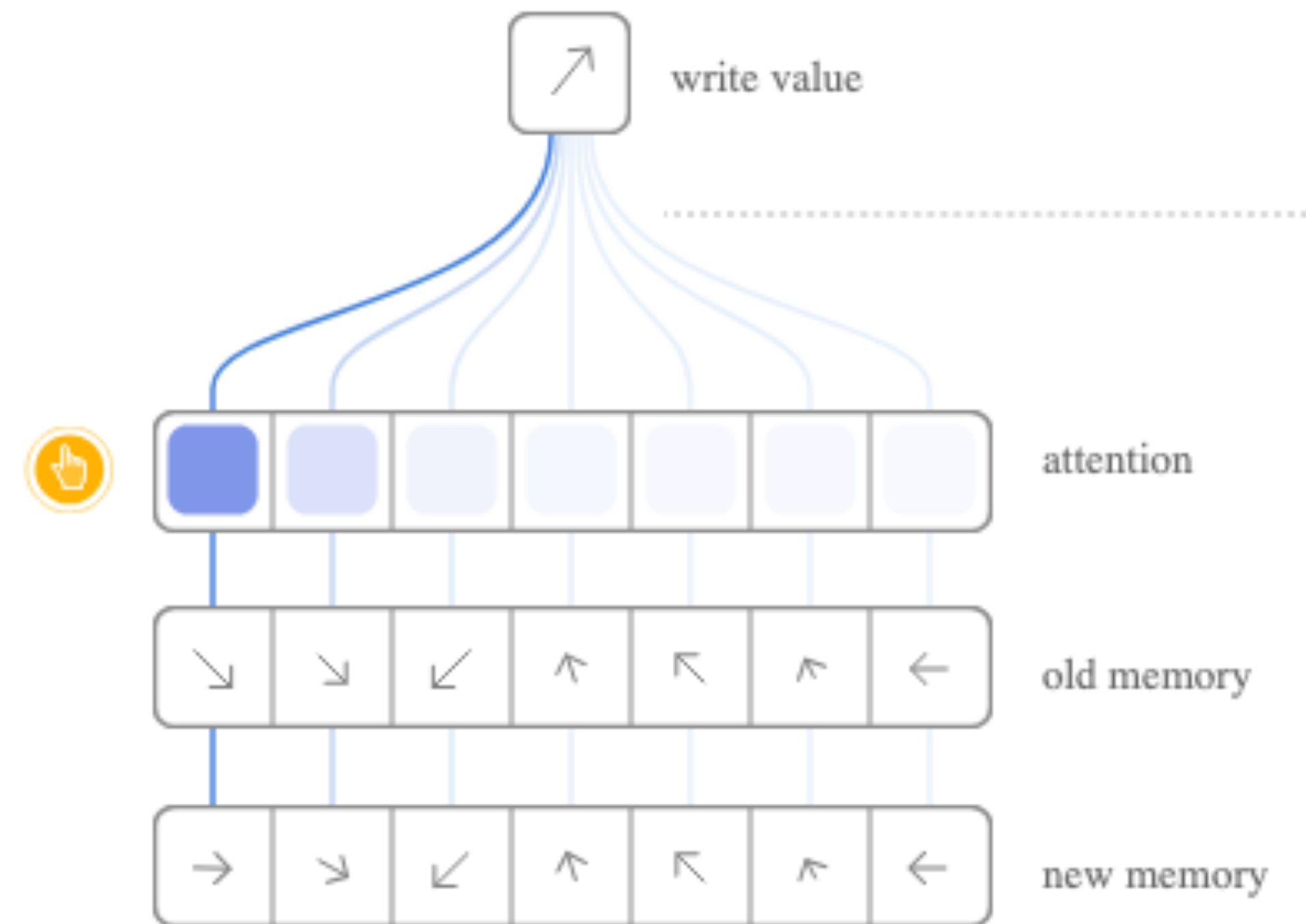
Controller와 Memory bank
= head와 tape

차이점은 바로!! soft-attention!

Neural Turing Machine

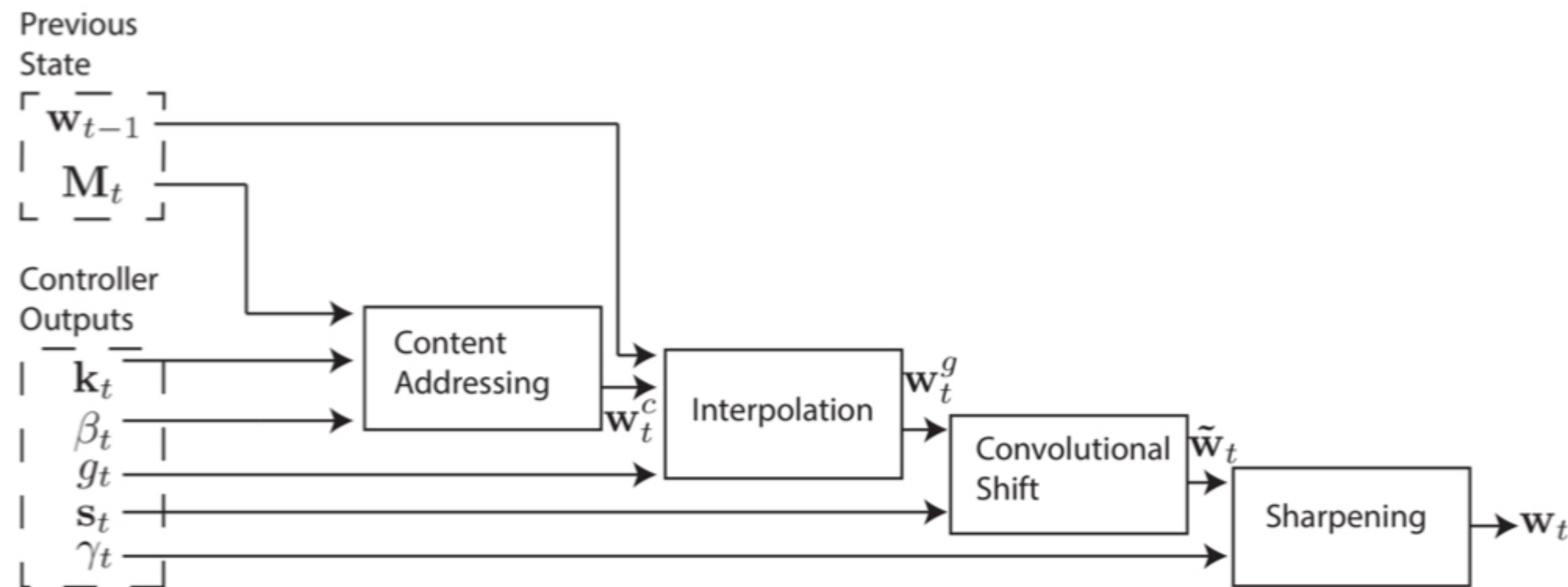


$$r \leftarrow \text{Sum } w(i)M(i)$$



$$M_{\text{new}}(i) = M (1 - w(i)e) + w(i)a$$

Neural Turing Machine - w?



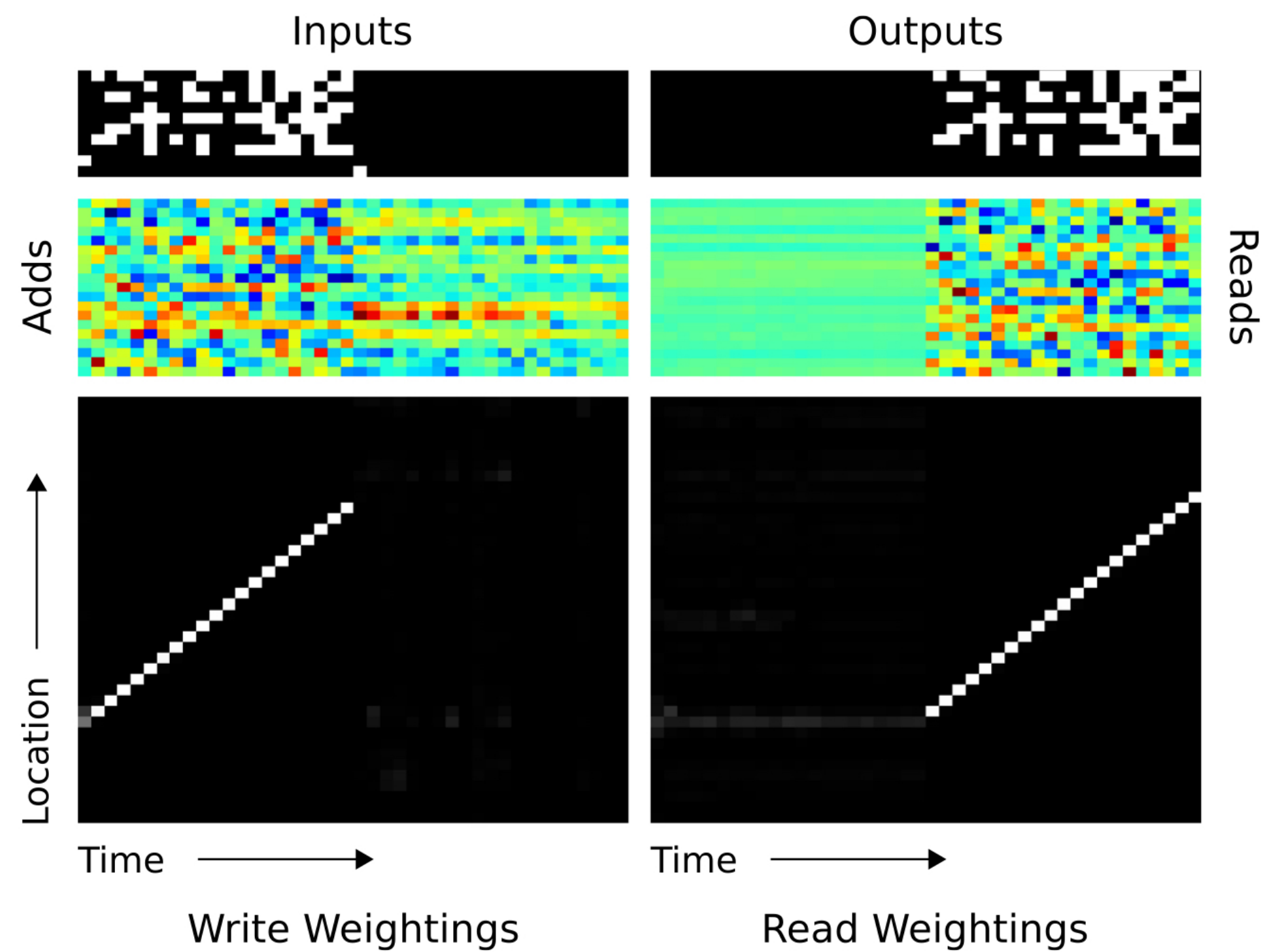
- Focusing by Content: Receive info with soft attention
- Focusing by Location: gate interpolation/shift scale

Neural Turing Machine - w?

- Focusing by Content: 당근 필요!
- Focusing by Location: 곱셈과 같은 연산에서 hard attention에 가까운 Inductive bias를 위해서

Neural Turing Machine!!

- it can do copy, repeat copy, associative recall, sorting!!



Neural Turing Machine

karpathy.github.io/2015/05/21/rnn-effectiveness/

further reading 부분에서 hard attention과 soft attention에 대한 언급

- non-differentiable in hard attention -> soft attention!!
- everything attended(but softly) in soft attention -> hard attention!!

more story

- non-differentiable in hard attention -> soft attention!!

hard attention utilizing REINFORCE trick

Cross-Entropy = bits for encoding distribution

-> trajectory score에 비례하여 학습 유도($f(x)\log p(x)$)

more story

- everything attended(but softly) in soft attention -> hard attention!!

soft attention = key vector interpolation with query information

query: this is important!!

soft attention: ok i'll give it more weights